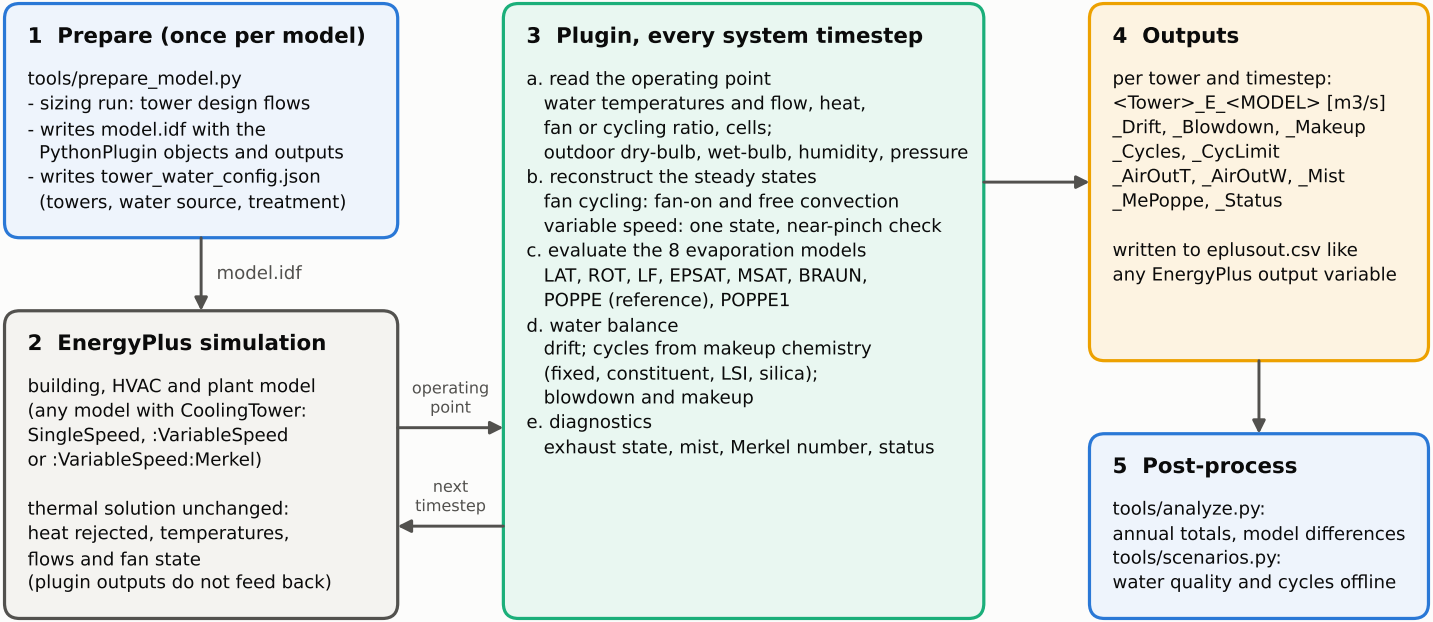
1 Prepare (once per model)
tools/prepare_model.py
- sizing run: tower design flows
- writes model.idf with the PythonPlugin objects and outputs
- writes tower_water_config.json (towers, water source, treatment)
model.idf
2 EnergyPlus simulation
building, HVAC and plant model (any model with CoolingTower: SingleSpeed, :VariableSpeed or :VariableSpeed:Merkel)
thermal solution unchanged: heat rejected, temperatures, flows and fan state (plugin outputs do not feed back)
operating point
next timestep
3 Plugin, every system timestep
a. read the operating point
water temperatures and flow, heat,
fan or cycling ratio, cells;
outdoor dry-bulb, wet-bulb, humidity, pressure
b. reconstruct the steady states
fan cycling: fan-on and free convection
variable speed: one state, near-pinch check
c. evaluate the 8 evaporation models
LAT, ROT, LF, EPSAT, MSAT, BRAUN,
POPPE (reference), POPPE1
d. water balance
drift; cycles from makeup chemistry
(fixed, constituent, LSI, silica);
blowdown and makeup
e. diagnostics
exhaust state, mist, Merkel number, status
4 Outputs
per tower and timestep:
<Tower>_E_<MODEL> [m3/s]
_Drift, _Blowdown, _Makeup
_Cycles, _CycLimit
_AirOutT, _AirOutW, _Mist
_MePoppe, _Status
written to eplusout.csv like any EnergyPlus output variable
5 Post-process
tools/analyze.py:
annual totals, model differences
tools/scenarios.py:
water quality and cycles offline